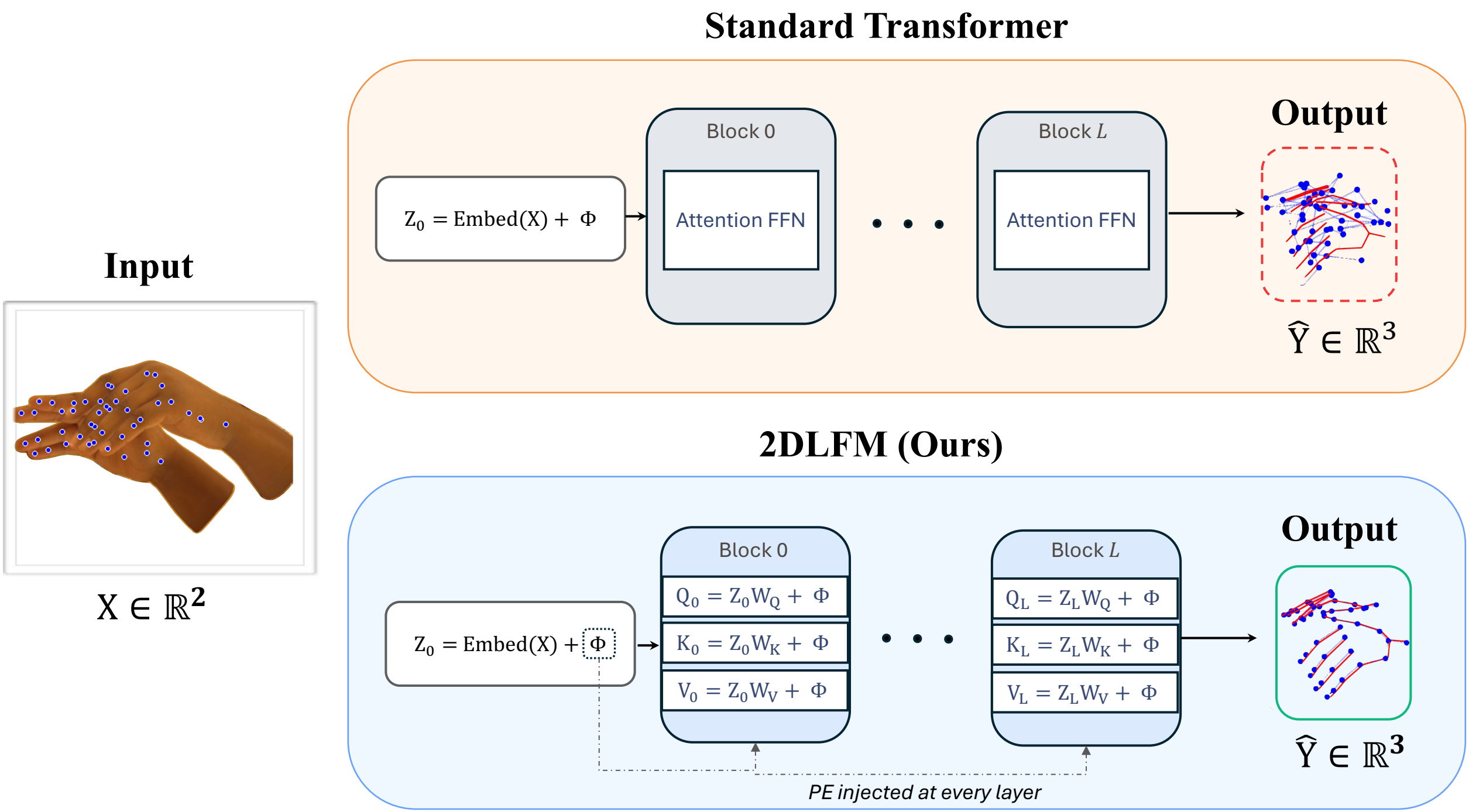
Standard Transformer
Input
$X \in \mathbb{R}^2$
$Z_0 = \text{Embed}(X) + \Phi$
Block 0
Attention FFN
Block L
Attention FFN
Output
$\hat{Y} \in \mathbb{R}^3$
2DLFM (Ours)
$Z_0 = \text{Embed}(X) + \Phi$
Block 0
$Q_0 = Z_0 W_Q + \Phi$
$K_0 = Z_0 W_K + \Phi$
$V_0 = Z_0 W_V + \Phi$
Block L
$Q_L = Z_L W_Q + \Phi$
$K_L = Z_L W_K + \Phi$
$V_L = Z_L W_V + \Phi$
Output
$\hat{Y} \in \mathbb{R}^3$
PE injected at every layer